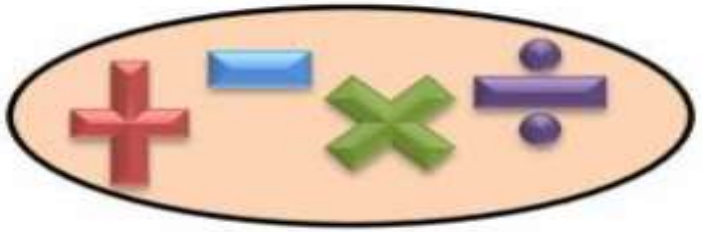
Data Handling





DATA Handling



Types of Data in Statistics

Primary Data



Secondary Data



Qualitative Vs Quantitative

Types of Data

- Quantitative
  - Discrete
  - Continuous
- Qualitative
  - Nominal
  - Ordinal

Qualitative Data

Definition: Qualitative data refers to non-numeric information that describes qualities, characteristics, or attributes of a subject. It captures the "what" or "how" of phenomena, focusing on descriptions rather than measurements.

Types of Qualitative Data:

1. **Nominal Data:**
    - Categorical data with no specific order or ranking.
    - Examples: Gender (Male/Female), Eye color (Blue, Brown).
2. **Ordinal Data:**
    - Categorical data with a defined order or ranking.
    - Examples: Customer satisfaction levels (Satisfied, Neutral, Dissatisfied).



Advantages of Qualitative Data:

1. In-depth Insights:
    - Provides detailed and nuanced understanding of topics.
2. Flexibility:
    - Methods like interviews and observations can be adapted to specific contexts.
3. Rich Contextual Information:
    - Captures the context and background of phenomena.
4. Useful for Exploratory Research:
    - Ideal for gaining initial understanding in new or complex areas.

Disadvantages of Qualitative Data:

1. **Subjectivity:**
    - Interpretation can vary, leading to potential bias.
2. **Time-Consuming:**
    - Data collection and analysis require significant time and effort.
3. **Limited Generalizability:**
    - Findings often apply to specific contexts and may not be universally applicable.
4. **Difficult to Analyze:**
    - Requires specialized techniques to identify themes and patterns.

Examples of Qualitative Data:

1. A teacher's observations of student behavior in a classroom.
2. Customer reviews of a product or service.
3. Transcripts from interviews conducted in a research study.
4. Photos of art installations in a gallery.
5. Audio recordings of focus group discussions.

Nominal Data

Definition: Nominal data is a type of qualitative data that represents categories or labels without any inherent order or ranking. It is used to classify data into distinct groups based on names, labels, or characteristics.

Characteristics of Nominal Data:

1. **No Order:**
   - Categories do not follow any sequence or ranking.
2. **Qualitative Nature:**
   - Represents descriptive information rather than numerical.
3. **Mutually Exclusive:**
   - Each category is distinct, and an observation can belong to only one category.
4. **No Arithmetic Operations:**
   - Cannot perform mathematical computations like addition or subtraction.

Real-World Examples of Nominal Data:

1. **Gender:** Male, Female, Non-Binary.
2. **Eye Color:** Blue, Green, Brown, Hazel.
3. **Marital Status:** Single, Married, Divorced, Widowed.
4. **Favorite Cuisine:** Italian, Mexican, Indian, Chinese.
5. **Geographical Regions:** Asia, Europe, Africa, America.

Ordinal Data is a type of categorical data that has a defined order or ranking, but the differences between the ranks are not necessarily equal or meaningful. It is used to represent data with a set order or sequence.

Characteristics of Ordinal Data:

1. **Ordered Categories**: The data can be arranged in a meaningful order, from lowest to highest or vice versa.
2. **Unequal Intervals**: The difference between the categories is not uniform or quantifiable.
3. **Qualitative Nature**: While ordinal data provides an order, it does not measure exact differences between categories.
4. **Non-Numeric or Numeric**: It can be represented either with numbers or labels, but the numbers don't represent exact values.
5. **Limited Operations**: Mathematical operations like addition or subtraction are not applicable to ordinal data, but ranking, median, and mode are.

Real-World Example:

- **Rating Scale:** In customer surveys, respondents might rate a product on a scale of 1 to 5, where:
  - 1 = Very Poor
  - 2 = Poor
  - 3 = Neutral
  - 4 = Good
  - 5 = Excellent The numbers reflect an order of preference but don't specify how much better "Good" is compared to "Neutral."

Quantitative Data

Definition: Quantitative data refers to numeric information that can be measured, counted, or expressed using numbers. It focuses on quantities and is often used for statistical analysis and mathematical computations.

Types of Quantitative Data:

1. **Discrete Data:**
    - Data that consists of distinct, separate values and cannot be divided further.
    - Examples: Number of students in a class, number of cars in a parking lot.
2. **Continuous Data:**
    - Data that can take any value within a given range.
    - Examples: Height of individuals, temperature measurements.

Advantages of Quantitative Data:

1. **Objectivity:**
    - Reduces bias as it relies on numeric measurements.
2. **Ease of Analysis:**
    - Can be analyzed using statistical tools to identify trends and patterns.
3. **Generalizability:**
    - Results are often applicable to larger populations.
4. **Precision:**
    - Provides specific, measurable insights.
5. **Replicability:**
    - Research can be repeated to verify results.

Disadvantages of Quantitative Data:

1. Lack of Depth:
    - Fails to capture underlying reasons, emotions, or motivations.
2. Context Ignored:
    - May overlook the context or background of the data.
3. Rigid Structure:
    - Data collection methods like surveys or experiments may limit flexibility.
4. Expensive and Time-Consuming:
    - Collecting large datasets may require significant resources.

Real-World Examples of Quantitative Data:

1. **Business:**
    - Monthly sales revenue of a company.
2. **Education:**
    - Test scores of students in a standardized exam.
3. **Healthcare:**
    - Blood pressure readings of patients over time.
4. **Sports:**
    - Number of goals scored by a team in a season.
5. **Environment:**
    - Daily rainfall measured in millimeters.

Discrete Data

Definition: Discrete data is a type of quantitative data that consists of distinct and separate values. These values are countable, finite, and cannot be divided into smaller parts that make sense in the context of measurement.

Characteristics of Discrete Data:

1. **Countable:**
   - Values can be counted as whole numbers.
   - Example: The number of students in a class.
2. **Finite Set of Values:**
   - Only specific values are possible, with gaps between them.
   - Example: The number of cars in a parking lot (1, 2, 3... but not 2.5).
3. **No Intermediate Values:**
   - Cannot have fractions or decimals between valid data points.
   - Example: You can't have 3.7 pets; you can only have 3 or 4.

Real-World Examples of Discrete Data:

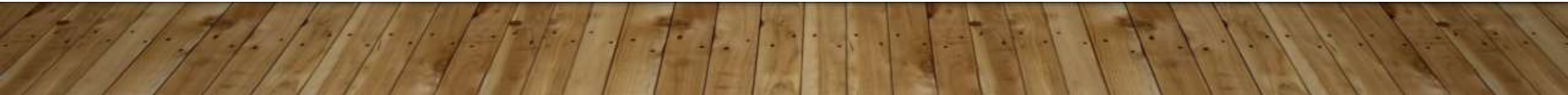
1. **Education:**
   - Number of students in a classroom.
2. **Transportation:**
   - Number of cars passing through a toll booth in an hour.
3. **Sports:**
   - Number of goals scored by a team in a match.
4. **Healthcare:**
   - Number of patients visiting a clinic daily.
5. **Business:**
   - Number of products sold by a store in a day.
6. **Technology:**
   - Number of software bugs reported in a system.

Continuous Data

Definition: Continuous data refers to quantitative data that can take any value within a range. It is measurable and can include fractions and decimals, representing values on a continuous scale.

Characteristics of Continuous Data:

1. **Infinite Possibilities:**
   - Can take any value within a specific range or interval.
2. **Divisible:**
   - Can be broken down into smaller units (e.g., 5.1, 5.12, 5.123).
3. **Quantitative Nature:**
   - Represents numerical measurements.
4. **Arithmetic Operations:**
   - Can perform mathematical operations like addition, subtraction, and averages.
5. **Measured, Not Counted:**
   - Derived through measurement tools like scales, thermometers, or clocks.



Real-World Examples of Continuous Data:

1. Healthcare:
    - Weight of a patient (e.g., 68.5 kg).
    - Blood pressure levels (e.g., 120.5 mmHg).
2. Science:
    - Temperature readings (e.g., 37.2°C).
    - Length of an object (e.g., 15.75 cm).
3. Economics:
    - Income levels (e.g., $50,000.75 per year).
    - Inflation rates (e.g., 2.5%).
4. Sports:
    - Running times (e.g., 9.58 seconds for 100m).
    - Distances jumped in long jump (e.g., 8.35 meters).
5. Environment:
    - Daily rainfall (e.g., 12.3 mm).
    - Air quality index (e.g., 42.7).



Data Types in Pandas

Pandas, a Python library for data manipulation and analysis, supports various data types for efficient storage and processing. These data types correspond to Python or NumPy types and are crucial for handling large datasets.

Common Data Types in Pandas:

1. **int64:**
    - Represents integer values.
    - Example: 1, 42, -7.
    - Usage: Storing counts, IDs, or other whole numbers.
2. **float64:**
    - Represents floating-point (decimal) numbers.
    - Example: 3.14, -0.001, 1.0.
    - Usage: Measurements, continuous variables, or ratios.



3. **object:**
    - Represents strings or mixed types.
    - Example: 'apple', '123', "Hello!".
    - Usage: Text data or columns with mixed data types.
4. **bool:**
    - Represents Boolean values (True/False).
    - Example: True, False.
    - Usage: Logical conditions, flags, or binary classifications.
5. **datetime64[ns]:**
    - Represents dates and times.
    - Example: '2023-01-01', '2025-12-31 10:15:00'.
    - Usage: Timestamps, scheduling, time-based analysis.

Why Data Types Matter in Pandas:

1. Performance Optimization:
    - Using appropriate types like `category` reduces memory usage.
2. Accurate Calculations:
    - Ensures proper handling of operations (e.g., arithmetic on `float` or `int`).
3. Error Prevention:
    - Avoids issues like string-based comparisons in numeric columns.
4. Compatibility:
    - Facilitates integration with other libraries or exporting to formats like CSV and Excel.

Data Splitting

Data splitting is a crucial step in machine learning and data analysis workflows. It involves dividing a dataset into separate subsets to train, validate, and test models effectively.

Why Data Splitting is Important:

1. **Prevent Overfitting:**
    - Ensures the model generalizes well to unseen data.
2. **Model Evaluation:**
    - Allows for unbiased evaluation of model performance.
3. **Hyperparameter Tuning:**
    - Provides a subset for optimizing model parameters without affecting testing data.



Common Data Splits:

1. **Training Set:**
    - Used to train the machine learning model.
    - Typically comprises 60-80% of the dataset.
2. **Validation Set:**
    - Used for model tuning and selection.
    - Often 10-20% of the dataset.
3. **Test Set:**
    - Used for final evaluation of the model's performance.
    - Usually 10-20% of the dataset.



Data Scaling

Definition:

Data scaling is like adjusting the heights of these students so they're all on the same level. It's a technique used in machine learning to transform data into a specific range or distribution. This makes it easier for algorithms to process and understand the information.

Why is it important?

- **Fair Comparisons:** Just like comparing student heights, scaling ensures that features with larger values don't unfairly dominate the learning process.
- **Improved Algorithm Performance:** Many machine learning algorithms work better when the data is scaled. This can lead to faster training times and more accurate predictions.
- **Preventing Numerical Issues:** Scaling can help prevent problems like numerical instability, which can occur when features have vastly different scales.

Common Scaling Techniques:

1. **Normalization:** This is like making everyone's height between 0 and 1. It's useful when you know the minimum and maximum values of your data.
2. **Standardization:** This is like giving everyone a "z-score." It transforms the data so that it has a mean of 0 and a standard deviation of 1. This is often preferred when the data doesn't have a clear upper or lower bound.

Real-World Example:

Let's say you're building a model to predict house prices. You have features like:

- **Size (square feet):** Can range from 500 to 5000
- **Number of bedrooms:** Usually between 1 and 5

Without scaling, the size feature would have a much larger impact on the model's predictions because of its larger range. Scaling ensures that both features contribute fairly to the model's decision.



Array Dimensions: Definition and Explanation

Definition: An array is a data structure that stores multiple values in a single variable. The dimension of an array refers to the number of indices or axes required to access elements within it. It helps in determining how the data is organized and accessed.

Types of Array Dimensions:

1. 1-D Array (One-Dimensional Array):
    - A 1-D array is a linear collection of elements, where each element is identified by a single index.
    - Example:

      `arr = [10, 20, 30, 40]`

      To access elements, we use a single index: `arr[0] = 10`, `arr[1] = 20`, etc.

2. **2-D Array (Two-Dimensional Array):**
   - A **2-D array** is like a table with rows and columns, where each element is identified by two indices (row and column).
   - **Example:**

     `arr = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`

     To access elements, we use two indices:

     `arr[0][1] = 2`, `arr[2][0] = 7`, etc.
3. **3-D Array (Three-Dimensional Array):**
   - A **3-D array** is like a collection of 2-D arrays (think of a stack of tables or matrices), where each element is identified by three indices.
   - **Example:**

     `arr = [[[1, 2], [3, 4]], [[5, 6], [7, 8]]]`

     To access elements, we use three indices:

     `arr[0][1][0] = 3`, `arr[1][0][1] = 6`, etc.

4. **N-Dimensional Array:**
    - An array can have any number of dimensions. In higher-dimensional arrays, we need more indices to access each element.
    - **Example:** A 4-D array might be used in scientific computing or image processing, where data is represented with four indices.

Characteristics:

- **Shape:** The shape of an array refers to the size of each dimension. For example, a 2-D array with 3 rows and 4 columns has a shape of (3, 4).
- **Indexing:** Arrays are indexed starting from 0 in most programming languages, meaning the first element is at index 0.

Real-World Example:

1. **1-D Array:**

   A list of temperatures for a week:

   ```
   temperatures = [30, 32, 35, 28, 29, 31, 33]
   ```

2. **2-D Array:**

   A table of student grades in subjects:

   ```
   grades = [[90, 85, 88], [76, 79, 80], [91, 94, 95]]
   ```

   This can represent grades for 3 students in 3 subjects.

3. **3-D Array:**

   An image stored in RGB (Red, Green, Blue) format with dimensions for height, width, and color channels:

   ```
   image = [[[255, 0, 0], [0, 255, 0]], [[0, 0, 255], [255, 255, 0]]]
   ```

   This represents a 2x2 pixel image with 3 color channels for each pixel.

1. Mean (Average):

The mean is the sum of all values in a dataset divided by the total number of values.

Formula:

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- $\sum X$ is the sum of all data points.
- N is the total number of data points.

Example:

For the data: [5, 10, 15, 20, 25]

$$\text{Mean} = \frac{5 + 10 + 15 + 20 + 25}{5} = \frac{75}{5} = 15$$

2. Mode:

The **mode** is the value that appears most frequently in a dataset. If no number is repeated, the dataset has no mode. If there are multiple numbers with the same highest frequency, there can be multiple modes.

Example:

For the data: [1, 2, 2, 3, 4, 4, 4, 5]

- The mode is **4** because it appears the most frequently (3 times).

3. **Median:**

The **median** is the middle value in a dataset when the values are arranged in ascending order. If there is an even number of data points, the median is the average of the two middle values.

Steps to find the median:

1. Arrange the data in ascending order.
2. If the number of data points (N) is odd, the median is the middle value.
3. If N is even, the median is the average of the two middle values.

Example:

For the data: [1, 3, 5, 7, 9]

- The median is **5**, the middle value.

For the data: [1, 2, 3, 4, 5, 6]

- The median is $\frac{3+4}{2} = 3.5$.

5. Variance:

The **variance** is the average of the squared differences from the mean. It is essentially the square of the standard deviation and provides a measure of the data's spread.

Formula:

$$\text{Variance} = \frac{\sum(X_i - \mu)^2}{N}$$

Example:

Using the same data [5, 10, 15, 20, 25] with mean $\mu = 15$:

1. Calculate the squared differences from the mean (same as for SD):
   - $(5 - 15)^2 = 100$
   - $(10 - 15)^2 = 25$
   - $(15 - 15)^2 = 0$
   - $(20 - 15)^2 = 25$
   - $(25 - 15)^2 = 100$
2. Find the average of these squared differences:

$$\frac{100 + 25 + 0 + 25 + 100}{5} = 50$$

So, the **variance** is **50**.

4. **Standard Deviation (SD):**

The **standard deviation** measures how spread out the values in a dataset are. A low standard deviation means the data points are close to the mean, while a high standard deviation means they are more spread out.

Formula:

$$\text{Standard Deviation} = \sqrt{\frac{\sum(X_i - \mu)^2}{N}}$$

Where:

- X_i is each data point.
- μ is the mean of the data.
- N is the number of data points.

Example:

For the data: [5, 10, 15, 20, 25], with mean $\mu = 15$:

1. Subtract the mean from each data point:
   - $(5 - 15)^2 = 100$
   - $(10 - 15)^2 = 25$
   - $(15 - 15)^2 = 0$
   - $(20 - 15)^2 = 25$
   - $(25 - 15)^2 = 100$
2. Calculate the average of these squared differences:

$$\frac{100 + 25 + 0 + 25 + 100}{5} = 50$$

3. Take the square root:

$$\text{SD} = \sqrt{50} \approx 7.07$$



Thank You